NeuTronics Analog Computer

Product Type: Educational / Experimental Analog Computing Platform
Institution: University Analog Project – NeuTronics Team
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Product Overview

The NeuTronics Analog Computer is a compact platform developed for educational and experimental use in real-time analog computation. It is capable of performing essential analog operations, including summation, subtraction, multiplication, differentiation, and integration.

Key Features

- Real-time analog computation
- Dual-channel input architecture
- Adjustable gain control
- Dual-supply powered design

Supported Analog Operations

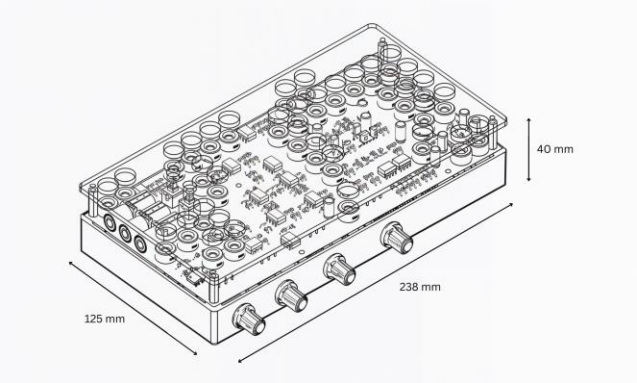
- Addition
- Subtraction
- Multiplication
- Integration
- Differentiation

Electronic Specifications

Parameter	Condition	Min	Typical	Max	Unit
Supply Voltage	Operating	±10	±12	±15	V
Input Voltage Range	Linear Operation	-10	-	+10	V
Output Voltage Range	Depend on supply	-10	-	+10	V
Bandwidth	-	10	15	-	kHz
Power Consumption	Nominal load	-	100	600	mW
Accuracy	Standard load	-	±2	±5	%
Noise Level	Standard load	-	5	100	mV
Settling Time	Step input	-	-	1	s
Linearity	Full-scale input	-	0.5	1	%

Physical Specifications

Parameter	Value	Unit
Height	40	mm
Width	125	mm
Length	238	mm
Weight	658	g



Interfaces and Controls

- Input Interfaces : Dual-channel analog inputs
- Output Interfaces : Analog output terminals
- Controls :
  - Gain adjustment
  - Operation mode selection

Operating Conditions

Parameter	Min	Typical	Max	Unit
Operating Temperature	-	25	30	°C

Notes

- All specifications are measured under standard laboratory conditions.
- Performance may vary depending on supply voltage and load conditions.
- Specifications are subject to change without prior notice.